

# How to build your own **green** observability pipeline



Presented by Diana Todea



# Whoami

HEAD OF DEVREL @VictoriaMetrics

CNCF Ambassador, TAG Lead Operational Resilience, OpenTelemetry Docs Approver

LinkedIn: <https://www.linkedin.com/in/diana-todea-b2a79968/>

GitHub: didiViking

Bluesky: @didiviking.bsky.social

X: @dianavtodea

Mastodon: @dianatodea

Instagram: @didivikingdevrel

Medium: @dianatodea



# The motivation

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Participate at a conference, saw a talk on [SPRUCE](#) project



Decided to fiddle with the project and integrate it with an observability backend and frontend



# The problem

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- 🌱 Cost and sustainability data often live in reports, not in engineering workflows
- 🌱 Observability has an environmental footprint
- 🌱 More data ≠ better observability



# The objective

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 Cloud cost is telemetry. Sustainability is telemetry

 We should be able to build viable Green Observability pipeline and observe our GreenOps data

## Building the pipeline

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 <https://github.com/didiViking/greenops-observability-reference-architecture>

 An open source green pipeline using Azure FOCUS, SPRUCE, Python, VictoriaMetrics, Grafana and Prometheus exposition format



# Architecture overview



# Why SPRUCE?

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 <https://github.com/digitalpebble/spruce>

 My project separates billing normalization from observability

 SPRUCE bridges that gap: it reads Azure FOCUS exports, normalizes the data into a FOCUS-compatible Parquet dataset and enriches each billing record with sustainability related attributes derived from open environmental models.

# Why VictoriaMetrics?

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🌱 VictoriaMetrics is an OSS project, which provides a lightweight, high-performance time-series database that is fully compatible with Prometheus ecosystem and requires very little operational overhead.



🌱 I'll use vmui (VictoriaMetrics UI) to:

- 🌱 discover metrics
- 🌱 inspect labels
- 🌱 experiment with PromQL
- 🌱 explore metric cardinality
- 🌱 validate the exported sustainability metrics

🌱 <https://github.com/VictoriaMetrics>

# The VictoriaMetrics Stack - Full Observability

Metrics + Logs + Traces: one consistent and efficient solution

## VictoriaMetrics

Metrics

Time series DB  
Prometheus-compatible  
MetricsQL

## VictoriaLogs

Logs

Fast logs database  
LogsQL query language  
Schema-less

## VictoriaTraces

Traces

Distributed tracing  
OpenTelemetry native  
10× compression

Unified on a single, coherent, cost-efficient platform · Open Source · Enterprise · Cloud



Simple, Reliable, Efficient Observability

[victoriametrics.com](https://victoriametrics.com)

# Execution step 1: Export billing data

## Export Azure FOCUS data

Cost Management: Virginia Diana Todea | Exports

billing account

Search

+ Create

Refresh

Run now

Enable

Disable

Delete

Feedback

Help

Overview

Change scope

Access control

Diagnose and solve problems

Reporting + analytics

Cost analysis

Exports

Scope: Virginia Diana Todea

Search to filter items...

You can now export to storage accounts with firewalls. [Learn more](#)

| Name                                       | Type                                     | Schedule status | Dataset version | Last run              | Frequency |
|--------------------------------------------|------------------------------------------|-----------------|-----------------|-----------------------|-----------|
| <input type="checkbox"/> Daily-actual-cost | Cost and usage details (actual)          | Active          | 2021-10-01      | 7/9/2026, 9:13 AM UTC | Daily     |
| <input type="checkbox"/> Daily-focus-cost  | Cost and usage details (FOCUS) - Preview | Active          | 1.0x2           | 7/9/2026, 5:13 AM UTC | Daily     |

## Execution step 2: Normalize and enrich with SPRUCE

 Execute SPRUCE against the generated input file

```
docker run \  
-i /data/focus.csv  
-f FOCUS
```

 SPRUCE produces a normalized Parquet dataset similar to

```
exports/cost-reports/result-focus
```

## Execution step 3: Convert data into metrics

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Install and run VictoriaMetrics, open vmui



Export Prometheus metrics (convert the SPRUCE Parquet dataset into Prometheus metrics)



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## Execution step 4: Store and query

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 Install and run Grafana OSS, configure Prometheus datasource

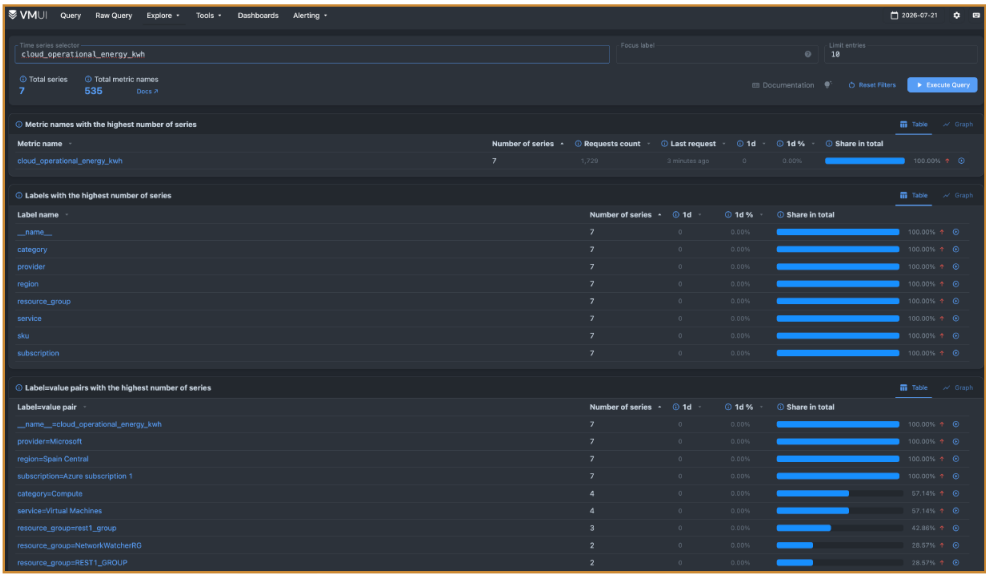
 Import Grafana dashboard using specific carbon related metrics



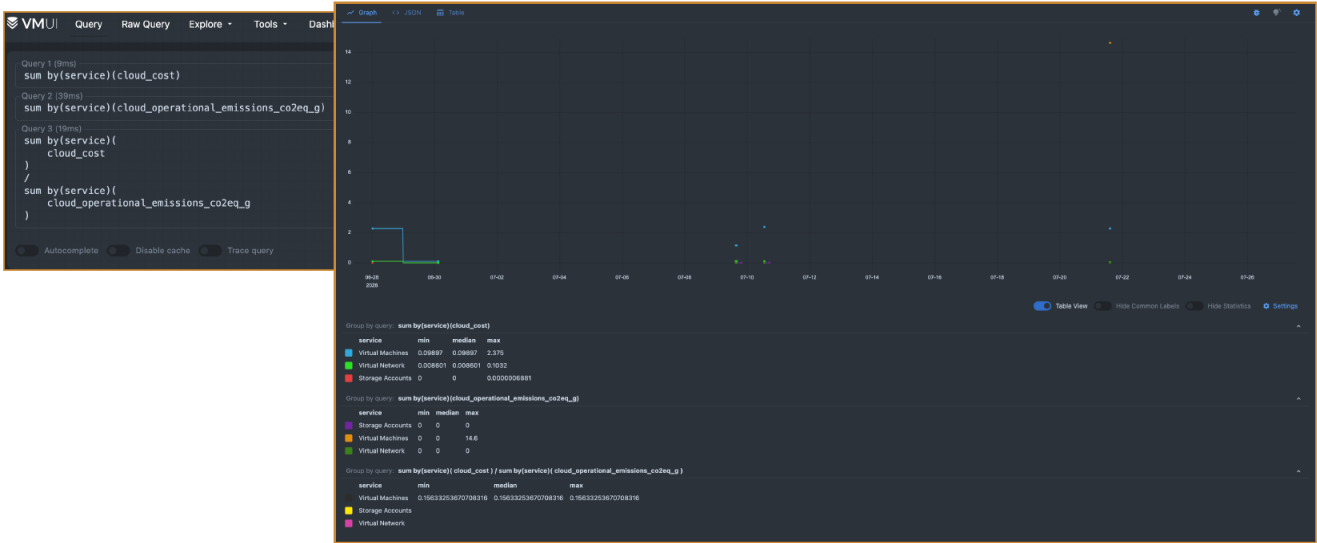
## Exported metrics

|    | Metric Name                           | Description                                                                             |
|----|---------------------------------------|-----------------------------------------------------------------------------------------|
| 1  | cloud_cost                            | Monetary cost for each billing record                                                   |
| 2  | cloud_usage_quantity                  | Resource consumption quantity                                                           |
| 3  | cloud_operational_energy_kwh          | Estimated operational energy consumption                                                |
| 4  | cloud_operational_emissions_co2eq_g   | Operational CO <sub>2</sub> emissions                                                   |
| 5  | cloud_embodied_emissions_co2eq_g      | Embodied carbon emissions                                                               |
| 6  | cloud_carbon_intensity                | Carbon intensity associated with the region                                             |
| 7  | cloud_power_usage_effectiveness       | Data center Power Usage Effectiveness (PUE)                                             |
| 8  | cloud_water_usage_effectiveness       | Water Usage Effectiveness (WUE)                                                         |
| 9  | cloud_water_cooling_l                 | Estimated cooling water consumption                                                     |
| 10 | cloud_water_electricity_production_l  | Water consumed during electricity production                                            |
| 11 | cloud_water_consumption_stress_area_l | Water consumption in stressed regions                                                   |
| 12 | cloud_resource_info                   | Inventory metric exposing cloud resources as Prometheus labels                          |
| 13 | cloud_meter_info                      | Inventory metric exposing Azure billing meters, categories, and subcategories as labels |

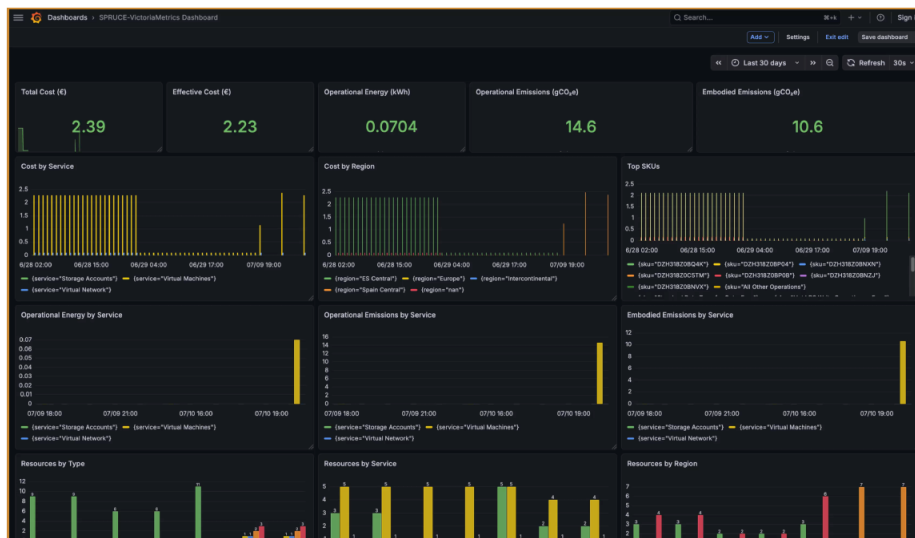
# Explore metrics in vmui



# Explore metrics in vmui



# Execution step 5: visualize your GreenO11y pipeline



## What didn't go as expected

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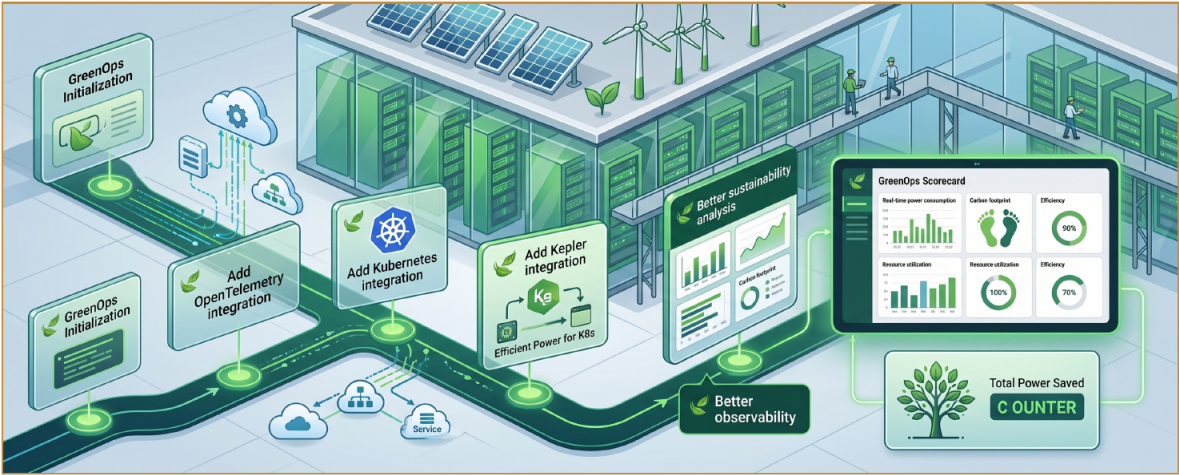
- 🌱 When I first created my pipeline, I created a data simulator, which added more complexity
- 🌱 I used Azure instead of AWS reports, due to some AWS free tier limitations
- 🌱 Found out that the Azure Billing had 2 types of reports and only the legacy one was supported by SPRUCE

## Working with maintainers

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- 🌱 Created a bug in SPRUCE project: <https://github.com/DigitalPebble/spruce/issues/246>
- 🌱 Collaboration with SPRUCE maintainer
- 🌱 Added new support for Azure FOCUS 1.0 and improve documentation

# Future improvements



## Take-aways

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- 🌱 Think about a concrete outcome and use case
- 🌱 Start simple and stick with OSS or cloud native tools (better collaboration and faster feedback)
- 🌱 Use a storage backend for your data and a frontend for visualization
- 🌱 If stuck, think if there are any alternatives already available

## Resources

1

<https://github.com/didiViking/greenops-observability-reference-architecture>

2

<https://github.com/digitalpebble/spruce>

3

<https://github.com/VictoriaMetrics>

4

<https://medium.com/@dianatodea/building-a-greenops-observability-pipeline-from-spruce-to-carbon-aware-metrics-with-open-source-c7e496dbc236>

5

<https://medium.com/@dianatodea/observability-has-a-data-hoarding-problem-70df85c88b9f>



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# Thank you!



Presented by Diana Todea

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